

Victoria Parrish

Location: East Lansing, MI

Email: vparrish@icecube.wisc.edu

POSITIONS AND EDUCATION

APPOINTMENTS

Michigan State University:

Postdoctoral Researcher in Experimental Particle Physics on the IceCube and P-ONE experiments
(2023 - present)

EDUCATION

University of Edinburgh PhD. in Particle Physics [Edinburgh, UK] Completion: June 1, 2023

Supervisor: Prof. Sinead Farrington

Topic: *Hunting exotics: the search for long-lived particles decaying to non-prompt tau leptons in the ATLAS detector and its essential technical developments*

Mount Holyoke College [S. Hadley, MA] Completion: May, 2019

B.A. with Honors in Physics

Advisor: Prof. Kevin McFarland, University of Rochester

Topic: *Z-Expansion Algorithm Efficiency in MINERvA CC0pi Neutrino Cross Section Data*

B.A. in Music

MEMBERSHIPS

United States Muon Collider Collaboration (USMCC) : 2025 - present

Pacific Ocean Neutrino Experiment (P-ONE) Collaboration : 2023 - present

Nobel Prize Winning IceCube Experiment Collaboration: 2023 - present

ATLAS Experiment Collaboration: 2019 - present

RESEARCH PROGRAM

Experimental particle physics research **across energy regimes**, from collider to astrophysical scales, integrating **detector design** with **machine learning-based software** and **data analyses** to **guide physics goals** into the future.

SELECTED PUBLICATIONS

For a full list of authored papers, please [view my Inspire HEP profile](#).

ANALYSIS

1. Collaboration, A. Search for light long-lived neutral particles that decay to collimated pairs of leptons or light hadrons in $\sqrt{s}=13$ TeV pp collisions at the ATLAS detector. J. High Energ. Phys. 2023, 153 (2023).
2. Aghaei, O. et al. Long Term Study of Sedimentation and Biofouling at Cascadia Basin, the Site of the Pacific Ocean Neutrino Experiment. Preprint at <https://doi.org/10.48550/arXiv.2507.09086> (2025).
3. Abbasi, R. et al. Sensitivity of the IceCube Upgrade to Atmospheric Neutrino Oscillations. in Proc. Sci. vol. 444 (Sissa Medialab Srl, 2024).
4. Collaboration, A. Search for displaced leptons in $\sqrt{s}=13$ TeV and 13.6 TeV pp collisions with the ATLAS detector. Phys. Rev. D 112, 012016 (2025).
5. Abbasi, R. et al. Search for Heavy Neutral Leptons with IceCube DeepCore. Preprint at <https://doi.org/10.48550/arXiv.2502.09454> (2025).
6. Collaboration, I. Physics potential of the IceCube Upgrade for atmospheric neutrino oscillations. Preprint at <https://doi.org/10.48550/arXiv.2509.13066> (2025).
7. *Good ν 's: neutrino detection at a Muon Collider Facility - in preparation*
8. *Search for long-lived particles decaying to displaced taus in Run 2 13 TeV pp collisions with the ATLAS detector - in preparation*
9. *Low energy tau neutrino flux measurements with the IceCube detector - in progress*

TECHNICAL SOFTWARE

1. Parrish, V. et al. Trigger and data-acquisition software developments for the Pacific Ocean Neutrino Experiment. in Proceedings of 39th International Cosmic Ray Conference — PoS(ICRC2025) vol. 501 1141 (Sissa Medialab, 2025).
2. Alimena, J. et al. Review of opportunities for new long-lived particle triggers in Run 3 of the Large Hadron Collider. Preprint at <https://doi.org/10.48550/arXiv.2110.14675> (2021).

3. Collaboration, A. The ATLAS Trigger System for LHC Run 3 and Trigger performance in 2022. J. Inst. 19, P06029 (2024).
4. *Offline and online methods for identifying displaced tau leptons in the ATLAS detector for Run 3 - in preparation*

TECHNICAL HARDWARE

1. Abbasi, R. & others. The LED calibration systems for the mDOM and D-Egg sensor modules of the IceCube Upgrade. (2025).
2. Fukami, S. Status and performance of mass-produced IceCube Upgrade mDOMs. Preprint at <https://doi.org/10.48550/arXiv.2507.07273> (2025).
3. Ghuman, D., Henningsen, F., Danninger, M. & on behalf of the P-ONE Collaboration. The Acoustic Calibration System for the Pacific Ocean Neutrino Experiment. in Proceedings of 38th International Cosmic Ray Conference — PoS(ICRC2023) vol. 444 1112 (SISSA Medialab, 2024).

COMMITTEES

- + Public communications and paper reviewer for HNL search at IceCube
- + Trigger coordinator and working group convenor for the P-ONE experiment
- + Chair of Particle Physics / BSM working group at Michigan State University
- + Chair of Events for the EDI committee at University of Edinburgh
- + TechFest and St. Margaret's School for Girls Women in STEM day event panel member

RELEVANT RESEARCH EXPERIENCE

TECHNICAL

IceCube - Northern Testing System lead at MSU (November 2023 - present)

- + Plan, configure, and execute hardware, firmware, and software setups for upcoming tests
- + Manage and train team of five personnel to populate lab 6 days/week

- + Communicate regularly to wider collaboration of time-sensitive testing results

P-ONE - trigger coordinator at MSU (November 2023 - present)

- + Evaluator of trigger chain development and primary investigator for trigger menu cohesion and deployment readiness
- + Representative of the wider trigger work for the P-ONE collaboration
- + Responsible for tracking project completion for team of eight undergraduates, PhDs, and Postdocs using Confluence, Jira, and fortnightly working group meetings

P-ONE - array performance studies lead at MSU (November 2023 - July 2025)

- + Head performance analyzer of machine-learning architecture options (RNN, GNN+RNN, CNN, transformers) for array-wide triggers
- + Supervisor of undergraduate student developing relevant simulation and architectures for training
- + Facilitator of weekly one-to-one meetings with students to ensure sustainable progress and project direction

IceCube - Local mDOM facilities and production supervisor at MSU (November 2023 - May 2024)

- + Organizer of lab setup, purchasing, and start-up personnel
- + Supervisor of three PhD students and overseer of the execution of lab tasks
- + Software maintainer of mDOM production and final acceptance testing facilities by debugging communications issues

ATLAS - tau/egamma expert on call at CERN, Switzerland at MSU (January 2022 - December 2022)

- + Designed pythonic framework to analyze event for Run reprocessing
- + Monitored online and offline trigger performance in sample reprocessings
- + Compiled, streamlined, and reported bugs and performances of tau, electron, and gamma triggers to wider collaboration

ATLAS control room run control and trigger shifter at CERN, Switzerland (January 2022 - December 2022)

- + Organized runs, and managed segments and resources for ATLAS during collisions
- + Monitored online triggers for ATLAS and debugged live issues during runs
- + Collaborated with sub-detector desks to design run configuration

ANALYSIS

IceCube - LLPs in-ice analysis lead (TrackGap) at MSU, Lansing, MI (November 2023 - present)

- + Technical supervisor of PhD student at the University of Uppsala
- + Project management of analysis, ensuring time-sensitive tasks are completed and presented to wider collaboration
- + Software developer of Icecube Pass 3 filter for muon tracks with 50m - 100m gap in detector signatures using a CNN

IceCube - Tau-neutrino (DoublePulse) analysis lead at MSU, Lansing, MI (June 2024 - present)

- + Designer of GNN and transformer reconstruction tools to search for Double Pulse, low-energy tau signatures in singularly triggered modules
- + Supervisor and project coordinator of analysis group of five searching for rare neutrino tau signatures in-ice
- + Software maintainer and structural designer of the analysis's git repository

ATLAS - PhD with University of Edinburgh, Edinburgh, UK (September 2019 - present)

- + ATLAS Run 2, release 21 long-lived particle to tau analysis
- + Design of Run 3 and release 22 studies of displaced tau trigger in ATLAS
- + Motivate discussion and team projects with tau long lived particle team

Fermilab - research undergrad with University of Rochester, Batavia, IL (June 2018 - August 2018)

- + Rebuilt reconstruction framework, GENIE, to account for new analytical model
- + Collaborated with MINERvA group to develop efficient reconstruction models for unaccounted data
- + Analyzed reconstruction versus data analytics to determine model and mode of best fit using NUISANCE

RELEVANT TEACHING EXPERIENCE

MSU- Course Organizer for Physics for Premed (December 2025)

- + Collaborate with biology and chemistry faculty to design physics content relevant to pre-med applications
- + Integrate concept inventories into class sessions to provide immediate feedback on student comprehension
- + Strengthen the flipped classroom model by scaffolding pre-class materials with short, clear, and purposeful content

MSU- Substitute lecturer for Electromagnetism for Physicists and Engineers (October 2024)

- + Engaged students by creatively adapting course material on capacitance by example of a theremin
- + Encouraged group discussion and collaboration for checkpoint problems
- + Ensured confused students were attended to and supported during lecture in a class of over 50 students

Edinburgh - Graduate TA for Electromagnetism, Thermodynamics and Data Analysis and Machine Learning (September 2019 - 2023)

- + Marker and demonstrator for courses for undergraduate and postgraduate students
- + Designed tutorial plans for two hour tutorial sessions
- + Communicated twice a week with lecturer to assist in preparing material

MHC - Peer-led Undergraduate Mentor (PLUM) in Introduction to Electromagnetism (September 2016- May 2019)

- + Facilitated group collaboration of students and between PLUMs
- + Marked 20+ Problem Sets a week
- + Organized weekly meetings with professor to discuss student problems/struggles

MHC - Electromagnetism Private Tutor (September 2017 - May 2018)

- + Assisted mature, disabled students with uniquely tailored problem solving frameworks
- + Advocated on behalf of student's needs to professor, dean, and counseling services in order to make a plan of action for success
- + Developed detailed, personalized, and topic specific set of notes and study packets for student's exams and assessments

RELEVANT EQUALITY, DIVERSITY, AND INCLUSION EXPERIENCE

IceCube Minority Mentor (September 2024 - present)

- + Designed and organized social justice book club; centering content of marginalized experiences in academia
- + One-on-one mentor to a masters student and two PhD students navigating queerness, disability, and race in academia
- + Promote structural changes within IceCube working groups to eliminate inequities

Demonstrator for St. Margaret's School for Girls Women in STEM day (February 2023)

- + Answered questions in outreach workshops on wind energy, gravity, and the night sky
- + Participated in panel discussions on PhD work at a middle school grade level
- + Collaborated with TechFest to set up and lead TechFest's Mobile Planetarium workshop

Chair of Events for the University of Edinburgh EDI Committee (September 2021 - May 2023)

- + Refactored the pedagogical and emotional intelligence workshop at Mount Holyoke College for faculty and TAs
- + Organized, hosted, and interviewed Dr. Donna Strickland, Nobel Prize Laureate in Physics for Women in STEM
- + Attended fortnightly EDI committee meetings to discuss plans for goals within the department

STEM with ME Founder (September 2020 - May 2021)

- + Lead sensitivity training based on Mount Holyoke College's pedagogical training materials
- + Hosted and facilitated monthly meetings with mentors to track progress of mentorship
- + One-on-one mentoring and college application prep of two high school girls looking to pursue a STEM degree

PANELS

- + *Invited speaker*: Women in Space and Science at Michigan State University (2026)
- + Graduate School in Europe at Mount Holyoke College (2024)
- + Under-represented gender luncheon with the IceCube Collaboration (2024)
- + Panelist for St. Margaret's School for Girls Women in STEM day (2023)
- + Student Representative EDI Town Hall at the University of Edinburgh (2021)
- + Graduate School for Women in STEM at Mount Holyoke College (2020)

CONFERENCES AND SEMINARS

TALKS

- + Oberlin College Physics Seminar Series: "Pioneering particle physics across energy regimes" (2025); *Invited Speaker*
- + Good neutrinos: Neutrino Physics at a Muon Collider: "Detecting TeV Neutrinos at a Muon Collider: Lessons from Cosmic Neutrino Experiments" (2025); *Invited Speaker*
- + Lepton-Photon: "P-ONE: first string developments, status, and outlook" (2025); *Invited P-ONE Collaboration Representative*
- + P-ONE Spring Collaboration Meeting: "Trigger and Data Acquisition for the P-ONE first string" (2025)
- + IceCube Spring Collaboration Meeting: "Reconstruction methods for invisible LLP searches in-ice" (2025)
- + P-ONE Fall Collaboration Meeting: **TDAQ convener** ; speaker "Trigger and Data Acquisition : Trigger design and implementation" (2024)
- + IceCube Collaboration: "Searching for invisible LLPs in-ice using CNNs" (2024)
- + IceCube Collaboration: **Plenary speaker** "Northern Test System : Cable Performance and Device Testing Configuration Results" (2024)
- + P-ONE Collaboration: "Trigger and Data Acquisition design report and review" (2024)
- + ATLAS UK: Speaker for analysis work (2023)
- + University of Edinburgh's Physics Postgraduate Association Seminar Series: "Long Lived Particles and Event Triggers at ATLAS" (2022, 2023)
- + ATLAS: TDAQ Week: "Review of tau triggers and their performances for Run 3" (2022)
- + SUPA Particle Physics Annual Theme Meeting: "Searching for displaced taus in ATLAS using RNNs" (2022)

POSTER SESSIONS

- + ICRC, Geneva Switzerland on behalf of the P-ONE Trigger and DAQ group (2025)

- + ATLAS UK (2021, 2022)
- + University of Edinburgh School of Physics and Astronomy Postgraduate Poster Session (2020)
- + Trigger Conference (2020)

WORKSHOPS

- + Science Outreach Weekend at Michigan State University (2023, 2024, 2025)
- + Emotional Intelligence for Faculty and Staff within the School of Physics and Astronomy at University of Edinburgh (2021, 2022, 2023)
- + Pedagogical practices and mentoring techniques for one-to-one sessions for STEM with ME (2019)
- + Secondary organizer of LLP Third Generation Theory workshop at University of Edinburgh (2019)

SKILLS AND LANGUAGES

HARDWARE

- + Expert in cable and hardware communications
- + Expert with oscilloscope operation
- + Expert with VNA measurements
- + Expert in soldering
- + Proficient with cable fabrication

SOFTWARE

- + Expert in data analysis and machine learning including:
 - o Expert in Tensorflow, Keras, and GraphNet
 - o Experienced in Hydra; Experienced in wandb
- + Languages:
 - o Expert in C++
 - o Expert in Python
 - o Expert in YAML
 - o Experienced in HTML
 - o Proficient in IDL
- + Operating Systems:
 - o Expert in UNIX, Windows, and Linux operating systems
- + Simulation and data analysis:
 - o Expert in GENIE
 - o Expert in NUISANCE

- Proficient in GEANT4
 - Experienced in Matlab and Mathematica
 - Expert in ROOT
 - Expert in IceTray
- + Organizational tools:
 - Expert with GitHub and GitLab

OTHER

- + Team leader and project management in groups of varying size and expertise
- + Ample leadership and collaboration experience
- + Clean room and electronics lab safety training
- + B2 Italian